

Runze Ma

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Education

Monash University	Master of Artificial Intelligence	2024.07 – 2026.11
Huazhong University of Science and Technology	B.Eng. in Automation	2020.09 – 2024.06

Publications

TRACE: Topology-Regularized Autoencoder for Interpretable Clinical ECG Diagnosis

- *NeurIPS 2026, CORE A*, Submitted, First Author (Co-Author)*
- We propose TRACE, a topology-regularized autoencoder that disentangles latent semantics into clinically meaningful subspaces via physiological constraints, achieving interpretable and robust ECG diagnosis.

Frozen Physics, Learned Residuals: Lightweight Hybrid Models for Low-Dimensional Dynamical Systems

- *AAAI 2027, CORE A*, Submitted, First Author*
- We propose a lightweight frozen-physics hybrid framework that keeps a known ODE as a backbone and learns only a small residual correction, achieving long-horizon stability with 5–18K parameters.

🔗 LiteMedCoT-VL: Parameter-Efficient Adaptation for Medical Visual Question Answering [\[link\]](#)

- *NLPCC 2026, Submitted, First Author*
- We propose LiteMedCoT-VL, a 2B vision-language model achieves 64.9% on medical VQA by distilling chain-of-thought reasoning from a 235B teacher via LoRA, outperforming models with twice the parameters.

BEAT-Net: Injecting Biomimetic Spatio-Temporal Priors for Interpretable ECG Diagnosis [\[link\]](#)

- *IEEE Journal of Biomedical and Health Informatics, JCR Q1, IF 7.7, Under Review, First Author*
- We propose BEAT-Net, a biomimetic framework mimicking cardiologist reasoning through QRS-centered tokenization and hierarchical spatio-temporal encoders for efficient, interpretable ECG diagnosis.

AEGIS-NET: Exploiting Physio-Kinetic Coupling Via Asymmetric Task-Aware Gating For Unified Biometrics, Activity Sensing, And Health Profiling [\[link\]](#)

- *IEEE Transactions on Consumer Electronics, JCR Q1, IF 9.9, Under Review, Second Author*
- We propose AEGIS-Net, a unified multi-modal framework for wearable devices that synergizes ECG, PPG, and IMU signals via an Asymmetric Task-Aware Gating mechanism.

🔗 Stacking-Enhanced Bagging Ensemble Learning for Breast Cancer Classification with CNN [\[link\]](#)

- *ICEEM 2023, Second Author*
- We propose BSECNN, a CNN-based ensemble using Bagging and Stacking for breast cancer classification.

🔗 REALM: Retrospective Encoder Alignment for LFP Modeling [\[link\]](#)

- *Journal of Neural Engineering, JCR Q2, IF 4, Under Review, Third Author*
- We propose REALM, a retrospective distillation framework that enables high-performance causal LFP decoding.

CPR: Causal Physiological Representation Learning for Robust ECG Analysis under Adversarial Perturbations [\[link\]](#)

- *NeurIPS 2026, CORE A*, Submitted, Fifth Author*
- We propose CPR, a robust diagnostic framework leveraging structural causal modeling to disentangle invariant pathological morphology from adversarial artifacts, achieving certified-level robustness and real-time inference efficiency.

Work Experience

NeuroML Lab at Tsinghua University *Research Assistant* 2026.05 – Present

- Developed and trained foundation models for biosignals.

Laboratory of Advanced Bioelectronics at SUAT [\[link\]](#) *Research Assistant* 2025.11 – Present

- Developed and trained multimodal models for biosignals.

Kingda Education *AI Technical Consultant* 2025.10 – 2026.06

- Delivered workplace AI training and provided technical consulting support for practical AI adoption.

- Designed and implemented task-oriented AI agents to streamline internal workflows.

Antalya Technology [\[link\]](#) *AI Product & Operations Lead* 2024.03 – Present

- Designed core prompts for **IELTS Master** [\[link\]](#), constructing a structured CoT framework to improve stability.
- Extracted user data via SQL and built automated Power BI dashboards to drive product iteration.

Juxin Financial Consulting *Data Analysis Intern* 2024.11 – 2025.02

- Integrated OCR with LLM APIs to automate invoice processing, boosting reconciliation efficiency by 50%.

 Project Experience

 **Wordmotion: One-Click Article-to-Video Pipeline with Remotion** 2026.06 – Present

- Built a AI agent skill that turns articles into narrated Remotion videos with word-level subtitles.
- Designed modular TSX slide components with frame-precise edge-tts audio-subtitle sync and multi-platform support.

 **Recipe Generation: From RNNs to RAG Menu Designer** 2026.04 – 2026.05

- Trained encoder-decoder RNNs with Bahdanau attention from scratch for ingredient-to-recipe generation.
- Fine-tuned T5 and GPT-2 with LoRA, and built a Menu Designer with RAG pipeline and LLM-as-Judge evaluation.

 **Multi-Agent RL for Autonomous Shuttle Delivery** 2025.05 – 2025.06

- Designed a tabular Q-learning MARL system on a 5×5 grid with collision detection and hybrid reward shaping.
- All four agents achieved 100% delivery success rate in focal agent evaluation across 16 start configurations.

 **Pac-Man AI: Classical Search & Adversarial Planning** 2025.03 – 2025.04

- Implemented A* search with domain-specific heuristics for single-goal, multi-corner, and full food collection pathfinding.
- Designed alpha-beta pruning with a handcrafted evaluation function for adversarial ghost avoidance.

 **Drone-Based Dual-Modal Vehicle Detection** 2023.11 – 2024.06

- Improved YOLO with RGB-IR feature fusion to resolve detection failures in complex lighting and weather.
- Validated on DroneVehicle dataset and optimized quantization for real-time embedded deployment.

2023 Summer Ethics of AI Program at North Carolina State University [\[link\]](#) 2023.07 – 2023.08

- Surveyed AI ethical controversies and built a multi-dimensional assessment framework.
- Authored a report proposing targeted ethical guidelines, receiving positive mentor feedback.

 **Pain Intensity Estimation via Spatio-Temporal Networks** 2023.02 – 2023.07

- Optimized feature extraction on UNBC-McMaster dataset by comparing multiple spatio-temporal architectures.
- Validated hybrid model advantages for non-invasive pain assessment via cross-validation.

Ensemble Learning with VAE for Breast Cancer Classification 2022.11 – 2023.02

- Proposed a Bagging & Stacking framework to address sample imbalance and improve reliability.
- Integrated VAE for data augmentation and feature selection to optimize classification performance.

 Certificates & Honors

2025 Smart Wearable and Sports Health Technology Challenge [\[link\]](#) - *National 1st Place* 2026.03

Scholarship for Scientific and Technological Innovation [\[link\]](#) - *1/30* 2022.09

Certificate of Professional Knowledge Assessment in Mathematical Modeling [\[link\]](#) 2022.08

Certificate of the University Artificial Intelligence Training Camp [\[link\]](#) 2022.08

MathorCup College Math Modeling Challenge [\[link\]](#) - *First Prize (Top 5%)* 2022.05

Mathematical Contest in Modeling [\[link\]](#) - *Honorable Mention (Top 20%)* 2022.02

Scholarship for Community Engagement [\[link\]](#) - *1/30* 2021.09

 Skills

- **Programming:** Python > C = Matlab > Java > C++
- **Software:** SQL, Power BI, VibeCoding
- **Languages:** Chinese (Native); English (IELTS 6.5); Arabic (Basic Spelling)